

A de Finetti endpoint for the growing- k mean-width question

Automated mathematics run `fa_banach_001`

Lane 14, candidate partial result

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Status. Candidate substantial partial result, likely valid pending expert review. For ordinary permutation-invariant k -extensions on $\mathbb{C}^d \otimes \mathbb{C}^d$, the packet proves

$$0 \leq w(\mathcal{E}_k) - w(\mathcal{S}_d) \leq C \min\{d^{-1}, d/k\}.$$

Thus $w(\mathcal{E}_{k(d)})/w(\mathcal{S}_d) \rightarrow 1$ whenever $k(d) \gg d^{5/2}$. The source’s general growing- k question, notably the intermediate window $d \lesssim k \lesssim d^{5/2}$, remains open.

1 Source question

The source is Cécilia Lancien, *k*-extendibility of high-dimensional bipartite quantum states, arXiv:1504.06459 [1]. In Section 9.1, PDF pages 17–18, it asks what can be said about the k -extendibility criterion on $\mathbb{C}^d \otimes \mathbb{C}^d$ when $k = k(d)$ grows with d . The exact prompt appears in Figure 1.

9. MISCELLANEOUS QUESTIONS

9.1. What about the mean width of the set of k -extendible states for “big” k ? All the statements proven so far, regarding either the k -extendibility of random-induced states or the mean width of the set of k -extendible states, converge towards the same (expected) conclusion: for any given $k \in \mathbf{N}$, the k -extendibility criterion becomes a very weak necessary condition for separability when the dimension of the considered bipartite system increases. So the natural question at that point is: what can be said about the k -extendibility criterion on $\mathbb{C}^d \otimes \mathbb{C}^d$ when $k \equiv k(d)$ is allowed to grow in some way with d ? Unfortunately, most of the results we established rely at some point on the assumption that k is a fixed parameter, and therefore do not seem to be directly generalizable to the case where k depends on d .

There is at least one estimate though that remains valid in this setting, which is the lower bound on the mean width of k -extendible states.

Theorem 9.1. *There exist positive constants $c_d \rightarrow_{d \rightarrow +\infty} 1$ such that, for any $k(d) \in \mathbf{N}$, the mean width of the set of $k(d)$ -extendible states on $\mathbb{C}^d \otimes \mathbb{C}^d$ satisfies*

$$w(\mathcal{E}_{k(d)}(\mathbb{C}^d; \mathbb{C}^d)) \geq c_d \frac{2}{\sqrt{k(d)}} \frac{1}{d}.$$

Proof. Let $d, k(d) \in \mathbf{N}$. For any $p \in \mathbf{N}$, the exact expression for the $2p$ -order moment established in Proposition D.1 of course remains true. So by the same arguments as in the proof of Proposition 2.3, we still have in that case

Figure 1: Section 9.1 of the source PDF: the growing- k question and the all- k lower bound.

Write $\mathcal{S}_d$ for the separable states on $\mathbb{C}^d \otimes \mathbb{C}^d$ and $\mathcal{E}_k(d)$ for the states admitting an extension to $\mathbb{C}^d \otimes (\mathbb{C}^d)^{\otimes k}$ that is invariant under permutations of the last k factors. This is the source’s ordinary permutation-invariant notion; the extension is not required to be supported on the Bose-symmetric subspace.

For a compact set $K \subset \mathcal{H}(\mathbb{C}^n)$, let

$$h_K(G) = \sup_{\rho \in K} \text{Tr}(G\rho), \quad w(K) = \frac{1}{\gamma(n)} \mathbb{E}_{G \sim \text{GUE}(n)} h_K(G), \quad \gamma(n) = \mathbb{E} \|G\|_2.$$

Lancien proves, uniformly for every $k = k(d)$,

$$w(\mathcal{E}_k(d)) \geq c_d \frac{2}{\sqrt{k}d}, \quad c_d \rightarrow 1, \quad (1)$$

and obtains a matching upper bound only in a slowly growing range. The known separable scale is

$$c d^{-3/2} \leq w(\mathcal{S}_d) \leq C d^{-3/2}, \quad (2)$$

by Aubrun and Szarek [3].

2 Proof intuition

The hierarchy converges to separability in trace norm by finite quantum de Finetti. Mean width is an average of support functions, and support functions are Lipschitz for trace norm with Lipschitz constant $\|G\|_\infty$ in direction G . Therefore a uniform trace-distance approximation of every k -extendible state by a separable state immediately becomes an additive mean-width estimate after averaging over G .

There are two dimension conversions. Finite de Finetti contributes d^2/k . A d^2 -dimensional GUE matrix has expected operator norm of order d , whereas the Gaussian normalization $\gamma(d^2)$ is of order d^2 . Their ratio contributes $1/d$. The resulting additive error is d/k . Dividing by the separable width $d^{-3/2}$ produces the relative error $d^{5/2}/k$.

The argument is robust but not expected to be sharp: it uses worst-case trace distance before averaging. It nevertheless supplies a complete far-large- k regime that is complementary to the source's small- k analysis.

3 Endpoint theorem

Theorem 1 (Uniform de Finetti endpoint). *There is a universal constant $C > 0$ such that, for every $d \geq 2$ and $k \geq 1$,*

$$0 \leq w(\mathcal{E}_k(d)) - w(\mathcal{S}_d) \leq C \min \left\{ \frac{1}{d}, \frac{d}{k} \right\}. \quad (3)$$

Consequently,

$$1 \leq \frac{w(\mathcal{E}_k(d))}{w(\mathcal{S}_d)} \leq 1 + C \frac{d^{5/2}}{k}. \quad (4)$$

In particular, for every integer-valued sequence $k = k(d)$ satisfying $k(d)/d^{5/2} \rightarrow \infty$,

$$\frac{w(\mathcal{E}_{k(d)}(d))}{w(\mathcal{S}_d)} \rightarrow 1.$$

Proof. Separable states have tensor-power extensions, hence $\mathcal{S}_d \subset \mathcal{E}_k(d)$. Monotonicity of support functions proves the left inequality in (3).

The quantitative finite quantum de Finetti theorem of Christandl, König, Mitchison, and Renner [2], in the relative-system form used for symmetric extensions, gives a universal C_0 such that for every $\rho \in \mathcal{E}_k(d)$ there is $\sigma_\rho \in \mathcal{S}_d$ with

$$\|\rho - \sigma_\rho\|_1 \leq C_0 \frac{d^2}{k}. \quad (5)$$

The source records the constant 2 in its unhalved Schatten-1 convention [1, Section 1]. Since the trace-norm diameter of the state space is 2, constants may be chosen so that

$$\|\rho - \sigma_\rho\|_1 \leq C_1 \min \left\{ 1, \frac{d^2}{k} \right\}. \quad (6)$$

Fix a Hermitian G on $\mathbb{C}^d \otimes \mathbb{C}^d$. Schatten Hölder duality and (6) give, for every $\rho \in \mathcal{E}_k(d)$,

$$\mathrm{Tr}(G\rho) \leq \mathrm{Tr}(G\sigma_\rho) + \|G\|_\infty \|\rho - \sigma_\rho\|_1 \leq h_{\mathcal{S}_d}(G) + C_1 \min \left\{ 1, \frac{d^2}{k} \right\} \|G\|_\infty.$$

Taking the supremum over ρ yields

$$0 \leq h_{\mathcal{E}_k(d)}(G) - h_{\mathcal{S}_d}(G) \leq C_1 \min \left\{ 1, \frac{d^2}{k} \right\} \|G\|_\infty. \quad (7)$$

For $G \sim \mathrm{GUE}(d^2)$, the standard Gaussian matrix estimates in the normalization used by Lancien give

$$\frac{\mathbb{E} \|G\|_\infty}{\gamma(d^2)} \leq \frac{C_2}{d}. \quad (8)$$

Indeed, the numerator is of order d and $\gamma(d^2) = \mathbb{E} \|G\|_2$ is of order d^2 . Averaging (7), dividing by $\gamma(d^2)$, and using (8) proves (3).

Finally, divide the Cd/k term in (3) by the lower bound $cd^{-3/2}$ in (2). This gives (4) after changing the universal constant, and the asserted limit follows. $\square$

Corollary 1 (Two rigorous phase endpoints). *For integer-valued $k = k(d)$:*

- (i) *if $k(d) = o(d)$, then $w(\mathcal{E}_{k(d)}(d))/w(\mathcal{S}_d) \rightarrow \infty$;*
- (ii) *if $k(d) \gg d^{5/2}$, then $w(\mathcal{E}_{k(d)}(d))/w(\mathcal{S}_d) \rightarrow 1$.*

Proof. Part (i) follows by dividing the source lower bound (1) by the upper bound in (2):

$$\frac{w(\mathcal{E}_{k(d)}(d))}{w(\mathcal{S}_d)} \geq c' \sqrt{\frac{d}{k(d)}} \rightarrow \infty.$$

Part (ii) is Theorem 1. $\square$

4 Upgrade audit and limitations

Eight focused routes were checked before promotion.

1. The source's exact Wick-moment formula was re-audited. Its defect-counting constraints reproduce the small- k barrier, and the missing uniform connection-coefficient estimate remains essential.
2. Monotonicity extends the source upper estimate to a plateau of order $(\log d)^{5/14}d^{-8/7}$, but this does not reach $d^{-3/2}$.
3. Trace-norm de Finetti plus Gaussian duality gives Theorem 1.
4. The ambient state-space width supplies the cap C/d in (3), without improving the endpoint exponent.
5. One-way-LOCC de Finetti estimates do not control general GUE support directions.
6. The stronger noisy-separability theorem of Navascués–Owari–Plenio applies to Bose-symmetric extensions, not to the ordinary permutation-invariant extensions defining $\mathcal{E}_k(d)$ here.
7. No Hilbert–Schmidt/reference-system theorem or legitimate Gaussian-cancellation argument with a better rate was found; the optimizer depends on G .

8. The August 2026 optimal argmax-rounding theorem was checked directly. Its ordinary two-sided bound at extension levels $(1, k)$ gives trace error $O(d/\sqrt{k})$ and hence width error $O(k^{-1/2})$. This improves d/k only where it is still above the separable scale, and does not improve the endpoint.

Thus the broad window

$$d \lesssim k \lesssim d^{5/2}$$

remains unresolved. Closing it appears to require either a de Finetti estimate tailored to average global Gaussian witnesses or a substantially more uniform analysis of Lancien’s correlated Gaussian operator norm.

Novelty check. The cheap run indexes and bounded searches through 11 August 2026 used the exact question wording, arXiv id, title, author, and combinations of *growing* k , *mean width*, *k-extendible*, and *finite de Finetti*. No explicit statement of (3) or of the $k \gg d^{5/2}$ sufficient regime was found. Novelty confidence is modest because the proof is a short, apparently unstated corollary of known estimates.

Human-review recommendation

Treat Theorem 1 as a likely valid substantial partial. The highest-value checks are: (i) the relative-system scope and trace-distance convention in (5); (ii) the d^2 -dimensional GUE normalization in (8); and (iii) whether provenance policy should describe the theorem as a new partial result or an unstated literature-implied corollary. The exponent and endpoint conclusion are insensitive to trace-distance factors of two.

References

- [1] C. Lancien, *k-extendibility of high-dimensional bipartite quantum states*, arXiv:1504.06459; published version dated 2016.
- [2] M. Christandl, R. König, G. Mitchison, and R. Renner, *One-and-a-half quantum de Finetti theorems*, Commun. Math. Phys. 273 (2007), 473–498; arXiv:quant-ph/0602130.
- [3] G. Aubrun and S. J. Szarek, *Tensor products of convex sets and the volume of separable states on N qudits*, Phys. Rev. A 73 (2006), 022109; arXiv:quant-ph/0503221.